# MACHINE LEARNING (ASSIGNMENT 03)

## Training and Evaluation of Supervised Machine Learning Models

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**GitHub Repository:** D:\ML Internship-ZeeOutsourcing\Machine Learning\Zee-Outsourcing-Internship-ML\Week-03-Supervised-Models

## 1. Introduction

The purpose of this assignment was to train and compare three machine learning classification models: K-Nearest Neighbors (KNN), Decision Tree, and Logistic Regression.

I used these models to predict whether a student might need academic support based on study habits, attendance, previous scores, and other factors. Their performance was compared using accuracy, precision, recall, F1-score, and confusion matrices.

## 2. Dataset Description

I used the Student Performance Factors dataset from Kaggle. It contains 6,607 records and 20 original columns, including numerical and categorical features.

**Dataset source:** https://www.kaggle.com/datasets/lainguyn123/student-performance-factors

The original target was Exam_Score. Since this assignment focuses on classification, I created a new target called Needs_Support:

- **Class 0:** Students scoring 65 or above (No Support).
- **Class 1:** Students scoring below 65 (Needs Support).

The cutoff of 65 was selected for this assignment.

The dataset contained 5,155 students (78.02%) in Class 0 and 1,452 students (21.98%) in Class 1. This shows that the dataset is imbalanced.

## 3. Data Preparation

First, I checked the dataset for missing values and duplicates. There were no duplicate records, but three categorical columns contained missing values.

I used median imputation for numerical features and most-frequent imputation for categorical features. OneHotEncoder was applied to convert categorical values into numerical features, while StandardScaler was used to scale numerical features.

The dataset was divided into 80% training and 20% testing data using random_state=42 and stratify=y.

- **Training records:** 5,285
- **Testing records:** 1,322
- **Features after preprocessing:** 42

I fitted the preprocessing steps only on training data to avoid data leakage. The original Exam_Score column was also excluded from the input features because the new target was directly derived from it.

## 4. Model Training

### K-Nearest Neighbors (KNN)

I initially trained KNN with K=5 and then tested odd K values from 1 to 15. K=7 and K=11 achieved the same highest validation accuracy of approximately 89.88%. I selected K=7 because it had a slightly better F1-score for Class 1.

### Decision Tree

I tested maximum depths of 2, 3, 5, 10, and None. Depth 10 achieved the highest validation accuracy of 87.51%. However, the deeper trees had much higher training accuracy without a similar improvement in validation accuracy, which indicated overfitting.

### Logistic Regression

I trained Logistic Regression on the processed dataset and used predict_proba() to examine the predicted class probabilities for five test students.

## 5. Model Comparison

| Model | Accuracy | Precision | Recall | F1-score |
|---|---|---|---|---|
| KNN | 87.97% | 84.74% | 55.33% | 66.94% |
| Decision Tree | 86.99% | 71.17% | 68.73% | 69.93% |

| | | | | |
|---|---|---|---|---|
| Logistic Regression | **98.26%** | **97.86%** | **94.16%** | **95.97%** |

Logistic Regression performed best among the three models. It achieved the highest accuracy and F1-score and identified more students who needed support.

## 6. Confusion Matrix Analysis

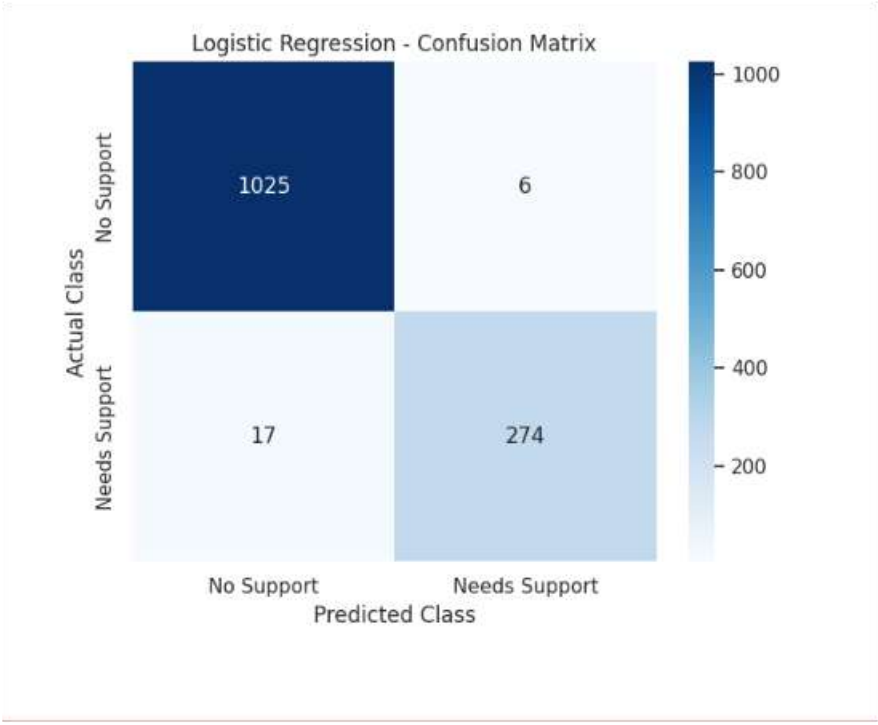



*Figure 1: Confusion Matrix of Logistic Regression*

The confusion matrix shows that Logistic Regression made the following predictions:

- **1,025 True Negatives:** Correctly predicted students who did not need support.
- **274 True Positives:** Correctly predicted students who needed support.
- **6 False Positives:** Students incorrectly predicted as needing support.
- **17 False Negatives:** Students who needed support but were incorrectly classified as not needing it.

Overall, the model made only 23 incorrect predictions out of 1,322 test records.

## 7. Most Important Evaluation Metric

For this problem, I consider **recall for Class 1** particularly important because missing a student who needs support is more concerning than recommending extra support to a student who may not need it.

Accuracy alone can be misleading because the dataset is imbalanced. For example, KNN achieved almost 88% accuracy, but its recall for students needing support was only 55.33%.

F1-score is also useful because it considers both precision and recall.

## 8. Error Analysis

I examined the incorrect predictions made by Logistic Regression to understand where the model struggled.

Out of 23 errors, 17 students had an exam score of 64 and four had a score of 65. The remaining two had scores of 80 and 82.

This means **91.3% of the errors occurred near the classification threshold of 65**. The model performed well overall but had difficulty classifying some students whose scores were close to the cutoff.

## 9. Challenges and Learning Outcomes

One challenge was selecting suitable values for KNN and Decision Tree. I learned that increasing model complexity does not always improve performance, as seen in the Decision Tree results.

Another challenge was evaluating models on an imbalanced dataset. I learned why accuracy should be considered alongside precision, recall, and F1-score.

This assignment helped me understand model training, hyperparameter tuning, performance evaluation, and the importance of avoiding data leakage.

## 6. Which Model Worked Best?

Logistic Regression gave the best results, achieving 98.26% accuracy and 95.97% F1-score for Class 1.

For this problem, recall is particularly important because missing a student who needs academic support is more concerning than recommending support unnecessarily. Logistic Regression achieved 94.16% recall, compared with 55.33% for KNN and 68.73% for Decision Tree.

Accuracy alone can also be misleading. Since approximately 78% of students belong to Class 0, a model predicting Class 0 for every student could still achieve 78% accuracy. KNN achieved nearly 88% accuracy but missed 130 of the 291 students who actually needed support.

## 7. Mistakes and What I Learned

I examined the 23 incorrect predictions made by Logistic Regression. Seventeen students scored 64, four scored 65, and the remaining two scored 80 and 82. Therefore, 21 out of 23 errors (91.3%) occurred near the classification threshold of 65.

The main challenges were preventing data leakage during preprocessing and selecting suitable hyperparameters without relying on test data. I also learned that increasing the depth of a Decision Tree can improve training accuracy without improving its performance on unseen data.

Comparing recall and F1-score helped me understand why accuracy alone is not sufficient for evaluating imbalanced classification problems.

## 10. Conclusion

Among the three models, Logistic Regression gave the best results, achieving **98.26% accuracy and 95.97% F1-score** for Class 1.

The results show that Logistic Regression was the most effective model for predicting the score-based academic support category in this dataset.